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MIL-M-38510/249A
13 February 1990
SUPERSEDING
MIL-M-38510/249
3 October 1988

MILITARY SPECIFICATION

MICROCIRCUITS, MEMORY, DIGITAL, CMOS, 1,048,576/1-BIT DYNAMIC RANDOM ACCESS MEMORY (DRAM), MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, dynamic, CMOS, 1,048,576/1-bit random access memory microcircuits. Two product assurance classes (S and B), four electrical performance categories and five case styles are provided and are reflected in the complete part number.

1.2 Part or identifying number (PIN). The part or identifying number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device types. The device types shall be as shown in the following:

Device type	Device organization	Temperature range (case)	Access/cycle time	Refresh interval
01	1,048,576/1-bit RAM	-55°C to +125°C	80/160 ns	8 ms
02	1,048,576/1-bit RAM	-55°C to +125°C	100/200 ns	8 ms
03	1,048,576/1-bit RAM	-55°C to +125°C	120/230 ns	8 ms
04	1,048,576/1-bit RAM	-55°C to +125°C	150/260 ns	8 ms

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
V	D-6 (18-lead, .960" x .310" x .200"), dual-in-line package
X	See figure 1 (20-lead, .640" x .420" x .090"), flat package
Y	See figure 1 (20-lead, .685" x .338" x .125"), J-leaded chip carrier package
Z	See figure 1 (20-terminal, .685" x .358" x .092"), rectangular chip carrier package
U	See figure 1 (20-lead, .680" x .385" x .095"), flat package

1.3 Absolute maximum ratings. 1/

Voltage on any pin relative to V_{SS}	-0.5 V to +7.0 V
Storage temperature range (ambient)	-65°C to +150°C
Power dissipation (minimum cycle time)	1.0 W
Lead temperature (soldering, 5 seconds)	+270°C
Junction temperature (T_J)	+175°C
Short circuit output current	50 mA
Thermal resistance, junction-to-case (θ_{JC}):	
Case V	See MIL-M-38510, appendix C

1/ Operation at maximum rating for extended periods of time will impair the reliability of the device.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441-5700, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5962

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1.4 Alpha particle induced soft error rate. The soft error rate due to alpha particles shall not exceed 9.16×10^{-11} errors per bit-day at sea level as defined in 4.4.3 and method 1032 of MIL-STD-883 and under the following conditions:

a. V_{CC} = 4.5 V dc.

b. T_C = +25°C.

c. Data pattern = checkerboard.

d. Cycle time = 500 ns.

1.5 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Power supply and signal reference (V_{SS})	0.0 V dc 2/
High-level input voltage (V_{IH}):	
All inputs	2.4 V dc minimum to V_{CC} + 1 V dc maximum
Low-level input voltage (V_{IL}):	
All inputs	-0.5 V dc minimum to +0.8 V dc maximum
Refresh cycle time (t_{RFSH})	8.0 ms
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

2/ V_{SS} is common for all supply voltages.

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3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Case outline. The case outline shall be in accordance with 1.2.3 and figure 2.

3.2.2 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.3 Functional block diagram. The functional block diagram shall be as specified on figure 3.

3.2.4 Truth table. The truth table shall be as specified on figure 4.

3.2.5 Die overcoat. Polyimide and silicone coatings are allowed as an overcoat on the die for alpha particle protection provided that each coated microcircuit inspection lot (reference MIL-M-38510, 3.1.3.8) shall be subjected to and pass the internal moisture content test (reference MIL-STD-883, test method 1018, 5000 ppm). The frequency of the internal water vapor testing may not be decreased unless approved by the preparing activity.

When a die overcoat is applied during wafer fabrication preventing inspection of the die structures beneath the overcoat, the requirements of 4.2a herein shall apply.

For those designs that use a die coat, a measurement of the thickness of the die coat shall be made on a sample of the qualification devices. This will then be used for the minimum limit for the measurement to be made in 4.4.4.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.5 herein.

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range of section 1.5 herein.

3.5 Electrical test requirements. Electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.5.1 Functional tests. The appendix contains a core set of test patterns to be used in conjunction with table III. These test patterns should be supplemented with test patterns implemented by the manufacturer that have been determined to stress sensitivities related to address and data pattern for the device to be supplied to this specification. These additional test patterns shall be defined during initial qualification and submitted to the qualifying activity for approval.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 46 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

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4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria apply:

a. When the alternate screening option of method 5004 of MIL-STD-883 is applied to class B, the following additional items are applicable:

(1) Burn-in duration for class B shall be 160 hours minimum.

(2) The following voltage stress test may be used to satisfy the alternate screening qualifications of method 5004 of MIL-STD-883. The voltage stress test shall be added to screening procedure after seal and before the interim electrical screen of method 5004. The voltage stress condition is shown on figure 5, (same as burn-in or equivalent), with the following modifications:

(a) V_{CC} = 7.0 V minimum.

(b) T_A = +125°C minimum for 12 hours minimum.

b. Burn-in test (method 1015 of MIL-STD-883);

(1) Static test using the circuit shown on figure 6 or equivalent. Ambient temperature (T_A) shall be +125°C minimum. Test duration for the static test shall be 48 hours minimum for class S. The 48-hour burn-in may, at the manufacturer's option, be broken into two sequences of 24 hours each (static I and static II) followed by interim electrical measurements.

(2) Dynamic test (test condition D or E) using the circuit shown on figure 5 or equivalent. Test duration and temperature shall be in accordance with method 5004 of MIL-STD-883.

c. Interim and final electrical tests shall be as specified in table II, except interim electrical tests prior to burn-in are optional at the discretion of the manufacturer.

d. The PDA shall be as specified in MIL-M-38510.

e. Those devices whose measured characteristics, after burn-in, exceed the specified delta (Δ) limits of table IV or electrical parameter limits specified in table III, subgroup 1, are defective and shall be removed from the lot. The verified failures divided by the total number of devices in the lot initially submitted to burn-in shall be used to determine the percent defective for the lot and the lot shall be accepted or rejected based on the specified PDA.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.3.1 Qualification extension. When authorized by the qualifying activity, for qualification inspection, if a manufacturer qualifies to a faster device type which is manufactured identically to a slower device type on this specification, the slower device type may be part I qualified without further qualification testing. At the manufacturer's request, the slower device types will be added to the QPL.

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TABLE I. Electrical performance characteristics.

Characteristic	Symbol	Conditions 1/ 2/ $-55°C \le T_C \le +125°C$, $4.5\ V \le V_{CC} \le 5.5\ V$	Device type	Limits		Unit
				Min	Max	
Supply current from V_{CC} (active)	I_{CC1}	$\overline{RAS}$ and $\overline{CAS}$ cycling t_{RALRAL} = 160 ns t_{RALRAL} = 200 ns t_{RALRAL} = 230 ns t_{RALRAL} = 260 ns	 01 02 03 04		 75 70 60 55	mA
Supply current from V_{CC} (standby)	I_{CC2}	$\overline{RAS}$ and $\overline{CAS}$ = V_{IH}	All		5	mA
Supply current from V_{CC} (refresh)	I_{CC3}	$\overline{RAS}$ cycling; $\overline{CAS}$ = V_{IH} t_{RALRAL} = 160 ns t_{RALRAL} = 200 ns t_{RALRAL} = 230 ns t_{RALRAL} = 260 ns	 01 02 03 04		 75 70 60 50	mA
Supply current from V_{CC} (page mode)	I_{CC4}	$\overline{RAS}$ = V_{IL}; $\overline{CAS}$ cycling t_{CALCAL} = 45 ns t_{CALCAL} = 55 ns t_{CALCAL} = 70 ns t_{CALCAL} = 80 ns	 01 02 03 04		 60 50 40 30	mA
Supply current from V_{CC} ($\overline{CAS}$ before $\overline{RAS}$ refresh)	I_{CC5}	$\overline{RAS}$ and $\overline{CAS}$ cycling See figure 7	 01 02 03 04		 75 70 60 50	mA
Supply current (latchup) due to I/O pin under/ over-voltage	I_{CC2L1}	See 4.4.1d In accordance with I_{CC2} I/O = V_{CC} +1 V for 10 ms I/O = -1 V for 10 ms	All		8	mA
Supply current (latchup) due to input under/over-shoot	I_{CC1L2}	See 4.4.1d In accordance with I_{CC1} Inputs transition V_{IL} = -2.0 V to V_{IH} = V_{CC} +2 V	 01 02 03 04		 70 75 65 60	mA
Supply current (latchup) due to I/O pin overvoltage during power up	I_{CC2L3}	See 4.4.1d In accordance with I_{CC2} All I/O = V_{CC} +1 during V_{CC} ramp from 0 V to 5 V	All		8	mA
Output high voltage	V_{OH}	I_{OH} = -5 mA, V_{CC} = 4.5 V V_{IH} = 2.4 V V_{IL} = 0.8 V	All	2.4		V

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions 1/ 2/ $-55°C \le T_C \le +125°C$ $4.5\ V \le V_{CC} \le 5.5\ V$	Device type	Limits		Unit
				Min	Max	
Output low voltage	V_{OL}	I_{OL} = 4 mA, V_{CC} = 4.5 V V_{IH} = 2.4 V V_{IL} = 0.8 V	All		0.4	V
Input leakage	I_{IH} I_{IL}	V_{IN} = 0.0 V to 6.5 V	All	-10	10	µA
Output leakage	I_{OZ}	V_{OUT} = 0.0 V to 6.5 V	All	-10	10	µA
Input capacitance address (A_0-A_9)	C_{IN1}	See 4.4.1c	All		7	pF
Input capacitance CLK ($\overline{RAS}$, $\overline{CAS}$, $\overline{WE}$, D_{IN})	C_{IN2}	See 4.4.1c	All		10	pF
Output capacitance	C_{OUT}	See 4.4.1c	All		8	pF
Functional test		V_{IL} = 0.8 V, verify output D_{OUT} V_{IH} = 2.4 V	All			
Random read or write cycle cycle time	t_{RALRAL}	See figures 7, 8, 9	01 02 03 04	160 200 230 260	10,000 10,000 10,000 10,000	ns
Read-modify-write or read-write cycle time	t_{RALRAL}	See figure 10	01 02 03 04	185 220 265 315	10,000 10,000 10,000 10,000	ns
Access time from $\overline{RAS}$ 3/	t_{RALQV}	See figures 8, 10, 11	01 02 03 04		80 100 120 150	ns
Access time from $\overline{CAS}$ 3/	t_{CALQV}	See figures 8, 10, 11, 12	01 02 03 04		20 25 30 40	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions 1/ 2/ $-55°C \le T_C \le +125°C$, $4.5\ V \le V_{CC} \le 5.5\ V$	Device type	Limits		Unit
				Min	Max	
$\overline{RAS}$ pulse width	t_{RALRAH}	See figures 7, 8, 9, 10	01	80	10,000	ns
			02	100	10,000	
			03	120	10,000	
			04	150	10,000	
$\overline{CAS}$ pulse width	t_{CALCAH}	See figures 7, 8, 9, 10, 12, 13	01	20	10,000	ns
			02	25	10,000	
			03	30	10,000	
			04	40	10,000	
$\overline{RAS}$ precharge time	t_{RAHRAL}	See figures 7, 8, 9, 10, 12, 13	01	70	10,000	ns
			02	80	10,000	
			03	90	10,000	
			04	100	10,000	
$\overline{CAS}$ to $\overline{RAS}$ precharge	t_{CAHRAL}	See figures 8, 9, 10, 13	All	10		ns
$\overline{RAS}$ to $\overline{CAS}$ delay	t_{RALCAL}	See figures 8, 9, 10, 12, 13	01	20	60	ns
			02	28	75	
			03	28	90	
			04	33	110	
$\overline{RAS}$ hold time	t_{CALRAH}	See figures 7, 8, 9, 10, 12, 13	01	25		ns
			02	25		
			03	30		
			04	40		
$\overline{CAS}$ hold time	t_{RALCAH}	See figures 8, 9, 10, 12, 13	01	80		ns
			02	100		
			03	120		
			04	150		
Row address setup time	t_{RAVRAL}	See figures 7, 8, 9, 10, 12, 13	All	3		ns
Row address hold time	t_{RALRAX}	See figures 7, 8, 9, 10, 12, 13	01	13		ns
			02	15		
			03	15		
			04	20		
Column address setup time	t_{CAVCAL}	See figures 7, 8, 9, 10, 12, 13	All	0		ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions 1/ 2/ $-55°C \le T_C \le +125°C$, $4.5\ V \le V_{CC} \le 5.5\ V$	Device type	Limits		Unit
				Min	Max	
Column address hold time	t_{CALCAX}	See figures 8, 9, 10, 12, 13	01 02 03 04	20 20 25 25		ns
Read command setup	t_{WHCAL}	See figures 8, 10, 12	All	0		ns
Read command hold (referenced to $\overline{CAS}$)	t_{CAHWX}	See figures 8, 10, 12	All	0		ns
Read command hold (referenced to $\overline{RAS}$)	t_{RAHWX}	See figures 8, 10, 12	01 02 03 04	10 10 10 10		ns
Output disable delay	t_{CAHQZ}	See figures 8, 10, 11, 12	All		35	ns
Write command setup	t_{WLCAL}	See figures 9, 13	All	0		ns
Write command hold (referenced to $\overline{WE}$)	t_{CALWX}	See figures 9, 13	01 02 03 04	15 20 25 30		ns
Write command hold (referenced to $\overline{RAS}$)	t_{RALWX}	See figures 9, 13	01 02 03 04	60 75 85 100		ns
Write pulse width	t_{WLWH}	See figures 9, 10, 13	01 02 03 04	15 20 25 30		ns
Write to $\overline{RAS}$ lead	t_{WLRAH}	See figures 9, 10, 13	01 02 03 04	25 25 30 40		ns
Write to $\overline{CAS}$ lead	t_{WLCAH}	See figures 9, 10, 13	01 02 03 04	20 25 30 40		ns
Data-in setup time	t_{DVCAL}	See figures 9, 10, 13	All	5		ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions 1/ 2/ $-55°C \le T_C \le +125°C$, $4.5\ V \le V_{CC} \le 5.5\ V$	Device type	Limits		Unit
				Min	Max	
Data-in hold time	t_{CALDX}	See figures 9, 10, 13	01 02 03 04	20 20 25 30		ns
$\overline{CAS}$ to write delay	t_{CALWL}	See figure 10	01 02 03 04	20 25 40 50		ns
$\overline{RAS}$ to write delay	t_{RALWL}	See figure 10	01 02 03 04	80 100 130 160		ns
Page mode cycle time	t_{CALCAL}	See figures 12, 13	01 02 03 04	45 55 70 80		ns
Page mode $\overline{CAS}$ precharge time	t_{CAHCAL}	See figures 12, 13	01 02 03 04	10 10 15 25		ns
Refresh period	t_{RFSH}		All		8	ms
$\overline{CAS}$ hold time $\overline{CAS}$ before $\overline{RAS}$ only	$t_{RALCAH1}$	See figure 7	All	30		ns
$\overline{CAS}$ setup time $\overline{CAS}$ before $\overline{RAS}$ only	t_{CALRAL}	See figure 7	01-03 04	10 15		ns ns
Column address hold time	t_{RALCAX}	See figures 8, 9, 10, 12, 13	01 02 03 04	60 70 85 100		ns
Data in hold time	t_{WLDX}	See figure 10	01 02 03 04	20 20 25 30		ns
Data in setup time	t_{DVWL}	See figure 10	All	0		ns

1/ Complete terminal conditions shall be as specified in table III.

2/ AC characteristics assume transition times (t_T) = 5 ns based on 0.8 V to 2.0 V input level swings. For testing to the opposite input level extremes (-1.0 V and 6.5 V), transition times of 10 ns may be used.

3/ Output load = I_{OH} or I_{OL} of table I and 50 pF minimum to V_{SS} or equivalent (see figure 11).

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TABLE II. Electrical test requirements.

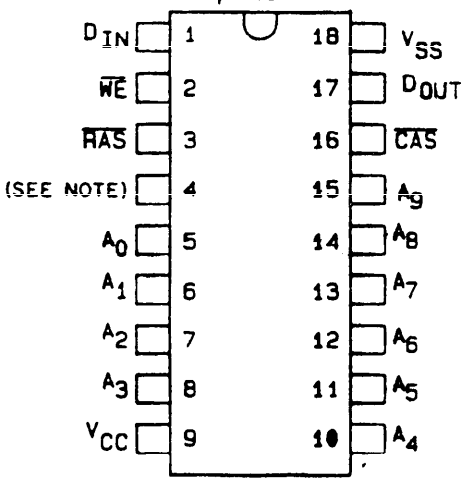
Line no.	MIL-STD-883 test requirements	Class S devices 1/ 2/				Class B devices 1/ 2/			
		Reference paragraph	Table III 3/ sub-groups	Table IV 4/ delta limits	Test circuit figure	Reference paragraph	Table III 3/ sub-groups	Table IV 4/ delta limits	Test circuit figure
1	Interim electrical parameters method 5004		1, 7				1, 7		
2	Static burn-in, method 1015	4.2b1 4.5.2	Required						
3	Same as line 1 5/		1*, 7*						
4	Dynamic burn-in, method 1015	4.2b2 4.5.2	Required		4	4.2b2 4.5.2	Required		4
5	Same as line 1	4.2	(1*, 7*)	Δ 6/		4.2e	(1*, 7*)	Δ 6/	
6	Final electrical parameters method 5004		1*, 2, 3, 7*, 8, 9, 10, 11				1*, 2, 3, 7*, 8, 9, 10, 11		
7	Group A test requirements, method 5005	4.4.1	1, 2, 3, 4, 7, 8, 9, 10, 11			4.4.1	1, 2, 3, 4, 7, 8, 9, 10, 11		
8	Group B test requirements, method 5005	4.4.2	1, 2, 3, 9, 10, 11	Δ					
9	Group C end-point electrical parameters method 5005					4.4.3	1, 2, 3, 9, 10, 11	Δ	
10	Group D end-point electrical parameters method 5005	4.4.4	1, 2, 3, 9, 10, 11			4.4.4	1, 2, 3, 9, 10, 11		

1/ Blank spaces indicate tests are not applicable.
2/ For subgroups 9, 10, and 11, only the worst value measured per device need be recorded when variables data is required (e.g., during qualification).
3/ (*) indicates PDA applies to subgroups 1 and 7 (see 4.2d).
4/ (Δ) indicates delta limit shall be required, and delta values shall be computed with reference to the previous interim electrical parameters (see 4.5.3).
5/ The device manufacturer may, at his option, delete line 3.
6/ The device manufacturer may, at his option, either perform delta measurements, or within 24 hours after burn-in (removal of temperature or bias), perform the final electrical parameters measurements, subgroup A1.

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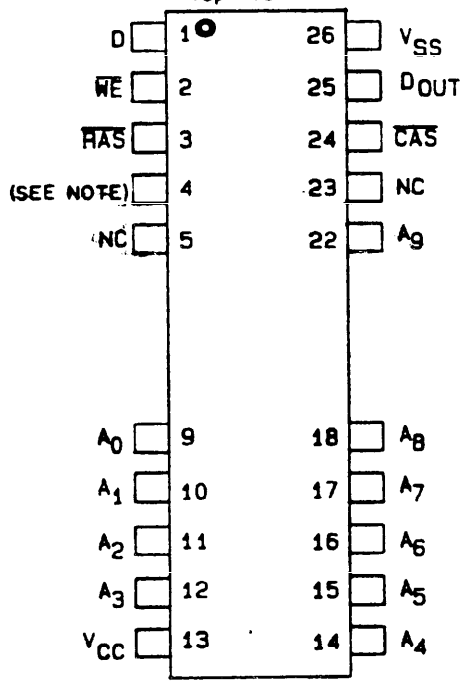
Case V

Top View

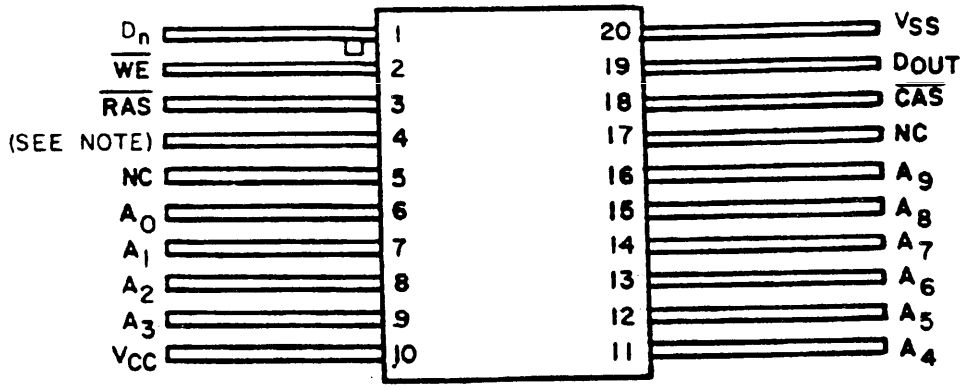


Cases Y and Z

Top View



Cases X and U

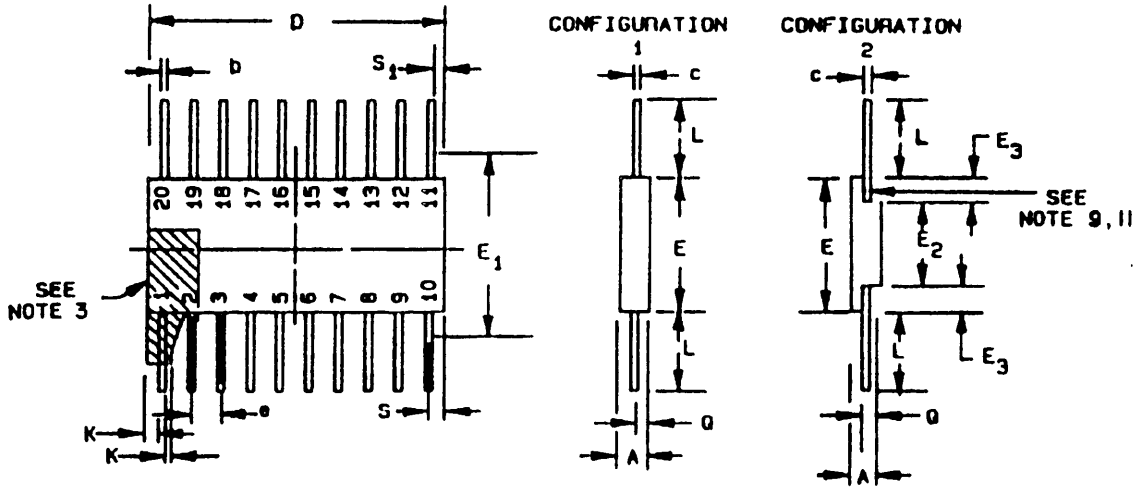


NOTE: Pins so marked are reserved for manufacturers test modes and should be tied to ground for proper device operation.

FIGURE 1. Terminal connections.

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Case X



Case X

Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.045	.105	1.14	2.67	---
b	.015	.019	0.38	0.48	7
c	.003	.006	0.08	0.15	7
D	---	.640	---	16.26	5
D_1	---	.530	---	13.46	5
E	.360	.420	9.14	10.67	---
E_1		.440	---	11.18	5
E_2	.180	---	4.57	---	---
E_3	.030	---	0.76	---	---
e	.050 BSC		1.27 BSC		6, 8
k	.008	.015	0.20	0.38	12
L	.250	.370	6.35	9.40	---
Q	.026	.040	0.66	1.02	4
S	---	.095	---	2.41	9
S_1	.005	---	0.13	---	10
α	.30°	90°	30°	90°	13

FIGURE 2. Case outlines.

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Case X

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Index area: A notch or a pin one identification mark shall be located adjacent to pin one and shall be within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark. Alternatively, a tab (dimension k) may be used to identify pin one. This tab may be located on either side as shown.
4. Dimension Q shall be measured at the point of exit of the lead from the body. Dimension Q shall be .0085 (0.215 mm) minimum when lead finish A is applied.
5. This dimension allows for off-center lid, meniscus, and glass overrun.
6. The basic pin spacing is .050 (1.27 mm) between centerlines. Each pin centerline shall be located with ±.005 (0.13 mm) of its exact longitudinal position relative to pins 1 and 20.
7. All leads: Increase maximum limit by .003 (0.08 mm) measured at the center of the flat, when lead finish A or B is applied.
8. Eighteen spaces.
9. Applies to all four corners (leads number 1, 10, 11, and 20 for configurations 1 and 2 and leads 2, 9, 12, and 19 for configuration 3).
10. Dimension S_1 may be .000 (0.00 mm) if leads number 1, 10, 11, and 20 for configurations 1 and 2 and leads number 2, 9, 12, and 19 for configuration 3 bend toward the cavity of the package within one lead's width from the point of entry of the lead into the body.
11. Optional configuration. If this configuration is used, no organic or polymeric materials shall be molded to the bottom of the package to cover the leads.
12. Optional, see note 3. If a pin one identification mark is used in addition to this tab, the minimum limit of dimension K does not apply.
13. Lead configuration is optional within dimension E, except when dimensions B and C apply.

FIGURE 2. Case outlines - Continued.

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Case Y

Inches	mm
.006	0.15
.010	0.25
.012	0.30
.022	0.56
.025	0.64
.028	0.71
.035	0.89
.044	1.12
.045	1.14
.048	1.22
.055	1.40
.056	1.42
.080	2.03
.102	2.59
.105	2.67
.125	3.17
.264	6.71
.308	7.82
.322	8.18
.338	8.59
.592	15.04
.608	15.44
.665	16.89
.685	17.40

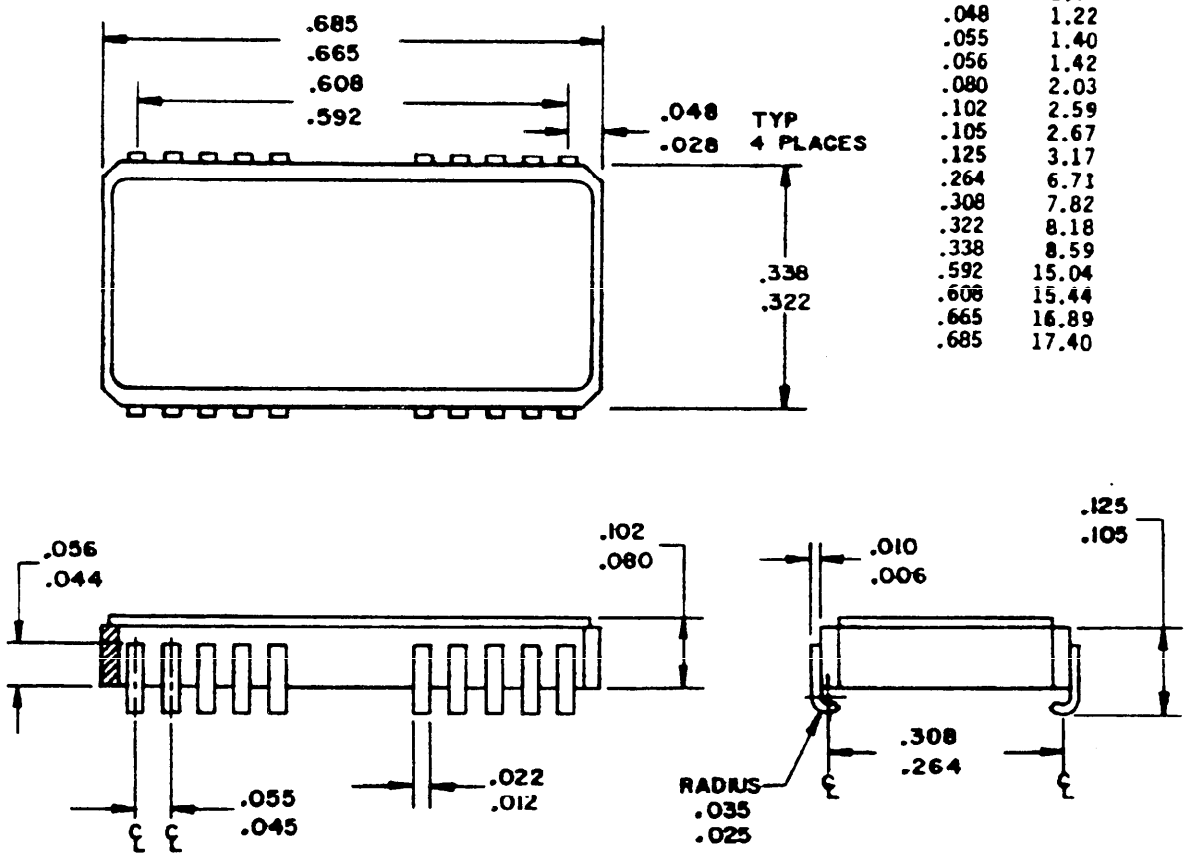
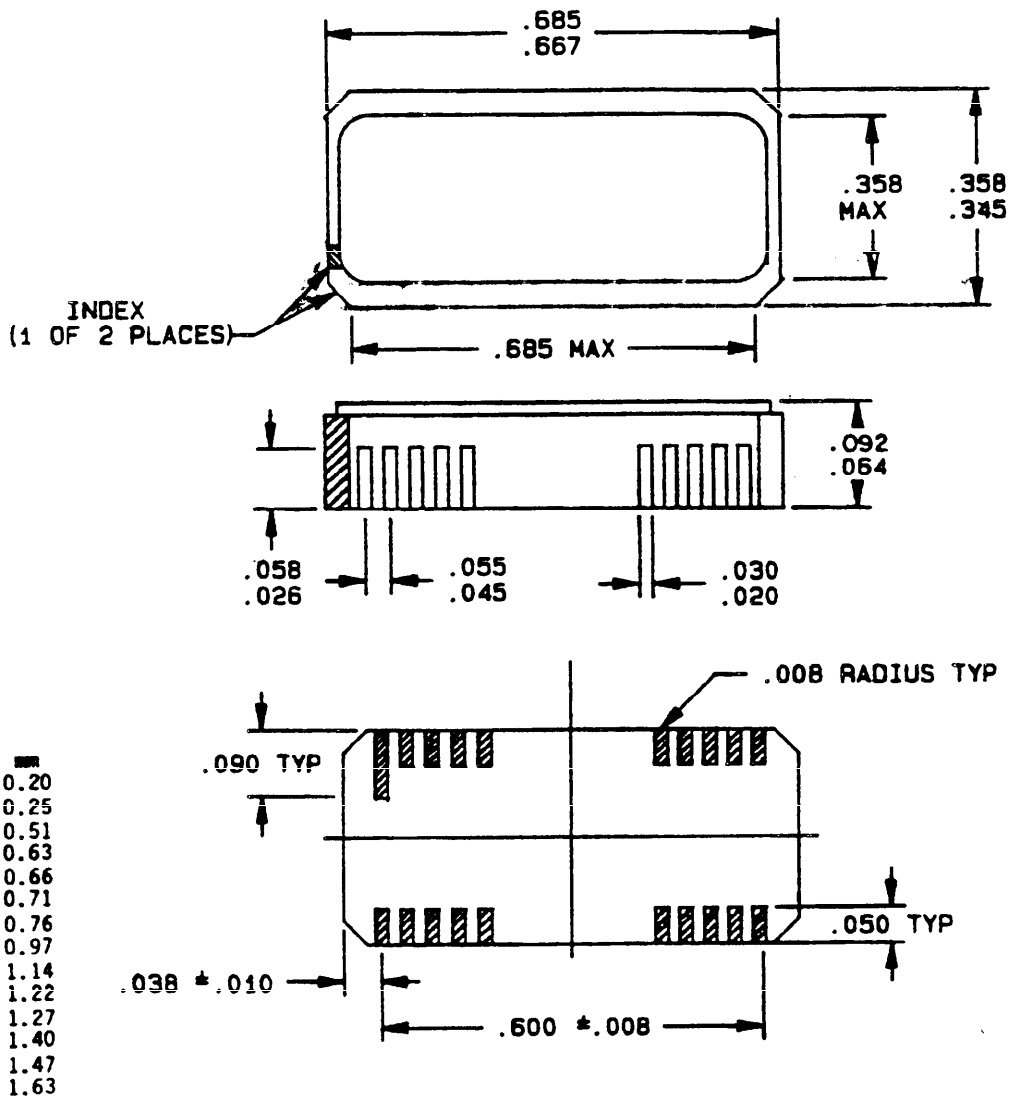


FIGURE 2. Case outlines - Continued.

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Case Z

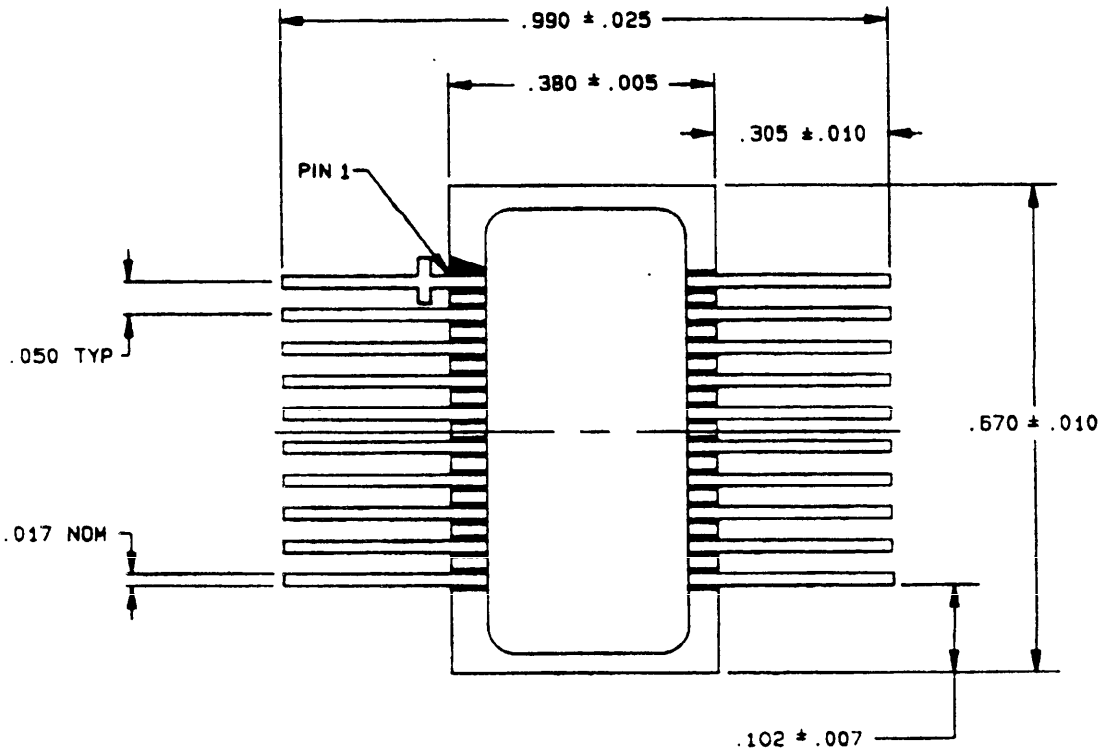


Inches	mm
.008	0.20
.010	0.25
.020	0.51
.025	0.63
.026	0.66
.028	0.71
.030	0.76
.038	0.97
.045	1.14
.048	1.22
.050	1.27
.055	1.40
.058	1.47
.064	1.63
.088	2.24
.090	2.29
.120	3.05
.300	7.62
.345	8.76
.358	9.09
.592	15.04
.600	15.24
.608	15.44
.667	16.94
.685	17.40

FIGURE 2. Case outlines - Continued.

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Case U



Inches	mm
.005	0.13
.007	0.18
.010	0.25
.012	0.30
.017	0.43
.025	0.63
.030	0.76
.050	1.27
.085	2.16
.305	7.75
.380	9.65
.670	17.02
.990	25.15

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Case outlines - Continued.

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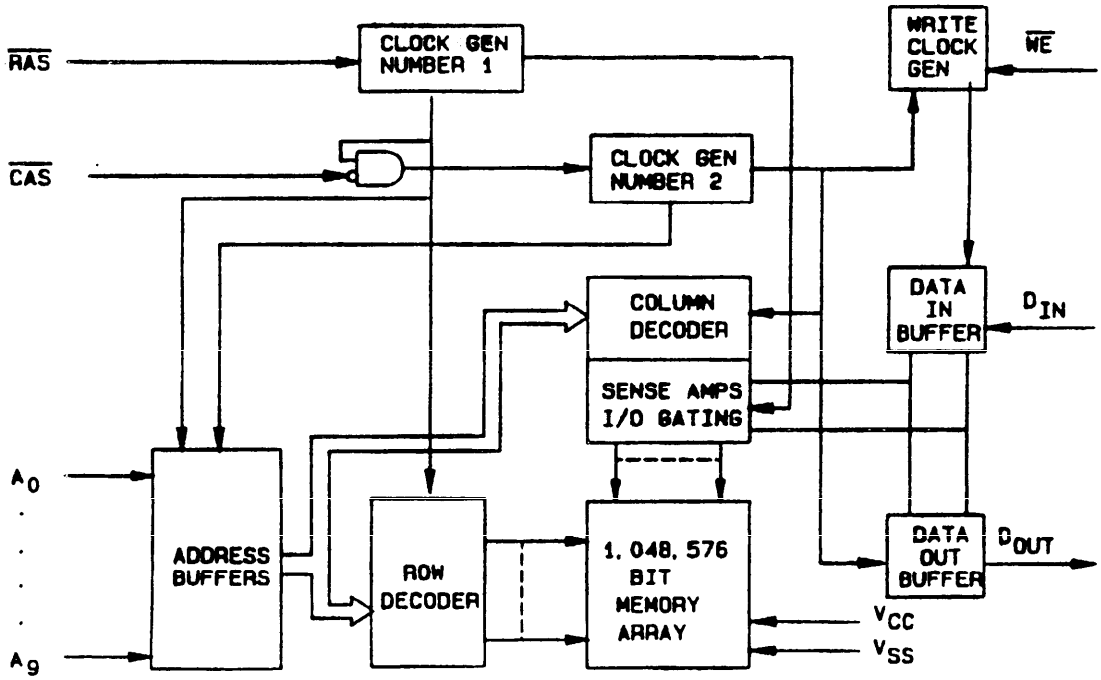


FIGURE 3. Functional block diagram.

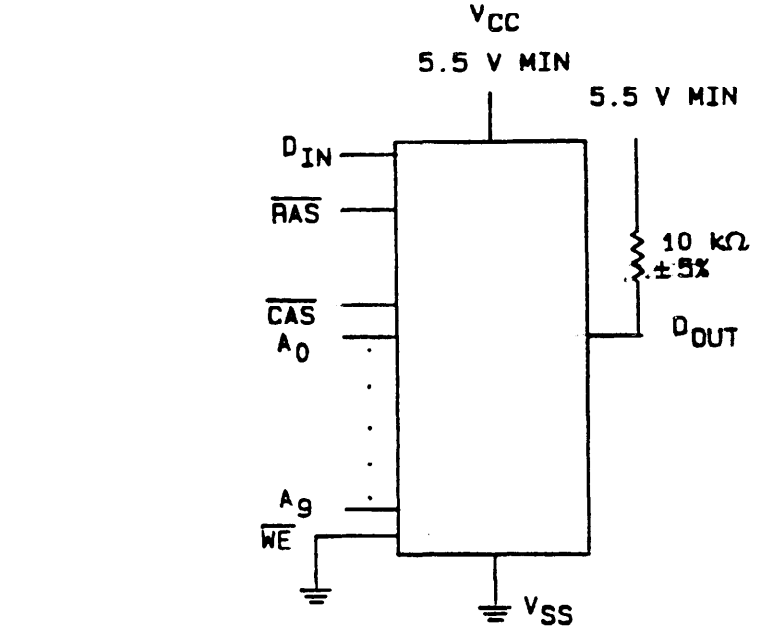
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:58Z
Check the source to verify that this is the current version before use.

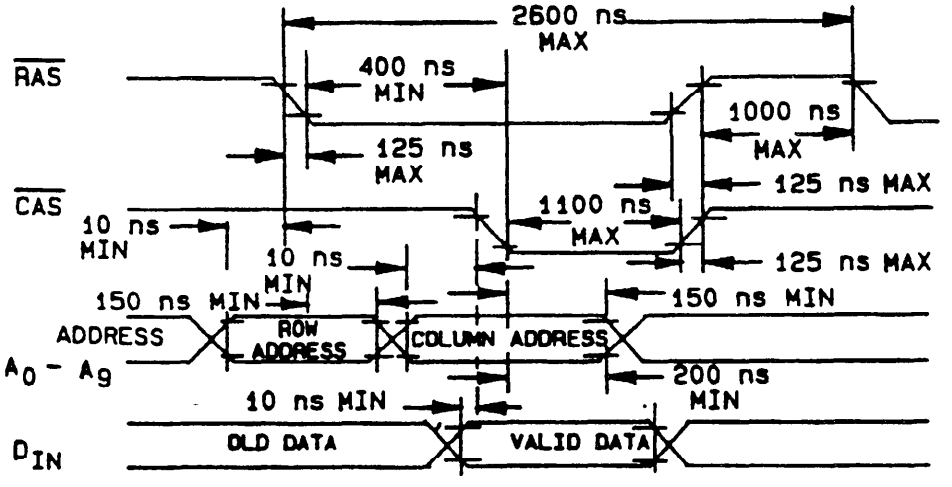
Operation (see note 1)	Inputs (see note 2)					Output
	$\overline{RAS}$	$\overline{CAS}$	D_{IN}	A_0-A_9	$\overline{WE}$	D_{OUT}
Deselected	H	H	X	X	X	High Z
Write 'D' to cell A(xy) (see note 3)	L	L	D (see note 4)	A(xy)	L	High Z (see note 5)
Read cell A(xy)	L	L	X	A(xy)	H	Data(xy)
$\overline{RAS}$-only-refresh	L	H	X	A(x) (see note 6)	X	High Z
Hidden refresh (see note 7)	L	L	X	A(x)	H	Data A(X-n, y-n)
Page mode write A(Y)	L	L	D	A(y)	L	High Z
Page mode read A(Y)	L	L	X	A(y)	H	Data A(y)
$\overline{CAS}$ before $\overline{RAS}$ refresh	L	L	X	X	X	High Z

NOTES:
1. A 100 µs pause is required after power-up, followed by eight $\overline{RAS}$ cycles before any truth table function.
2. 'X' = valid high or low.
3. A(xy) denotes proper address logic to address cell A(xy).
4. D denotes logic level for data to be written.
5. For early write timing, data-out remains at high impedance. For late write timing, data out is valid from access time until $\overline{CAS}$ goes to a high level.
6. A(x) depends only on A_0-A_8: A_9 is valid high or low.
7. When $\overline{CAS}$ = V_{IL} the data output will contain data from the last valid read cycle (i.e., N cycles before).

FIGURE 4. Truth table.

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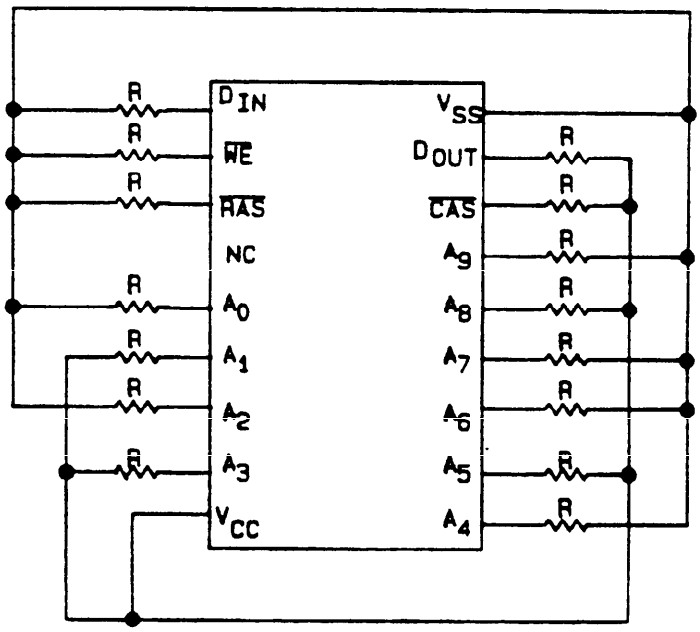




NOTES:
1. Address (A_0-A_9) = Binary count (LSB to MSB) every other cycle. (i.e., 0, 0, 1, 1, 2; .....1048574, 1048575, 1048575).
2. $\overline{WE}$ = Alternate valid high to valid low every other cycle. (i.e., write/read/write/read.....).
3. D_{OUT} = Option 1 or 3 from figure 11.

FIGURE 5. Burn-in and steady-state life test circuit.

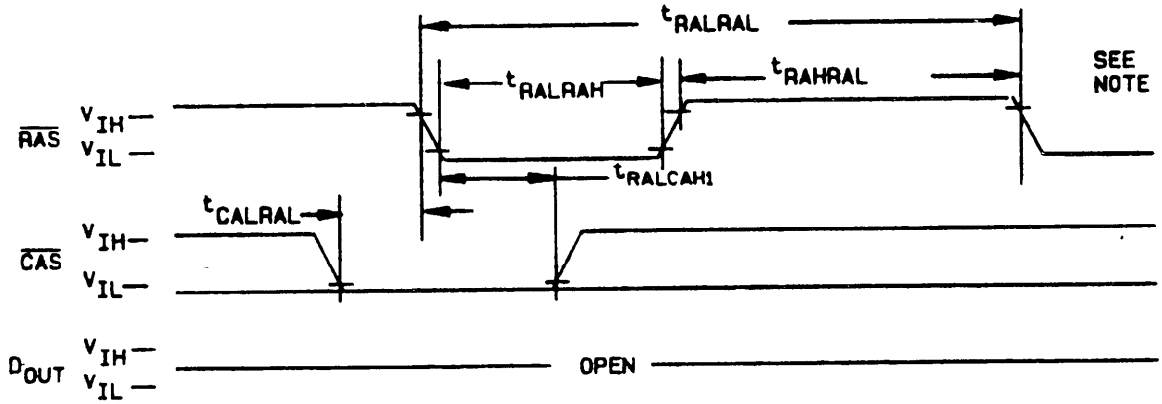
Source: https://assist.dla.mil -- Downloaded: 202610-21T13:58Z
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NOTES:
1. R = 1 kΩ ±5%.
2. V_{CC} = 5.5 V minimum.
3. For static II burn-in, reverse all input connections (i.e V_{CC} to V_{SS}).

FIGURE 6. Static burn-in circuit.

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NOTE: A_0 - A_9 = don't care (X).

FIGURE 7. CAS before RAS refresh cycle.

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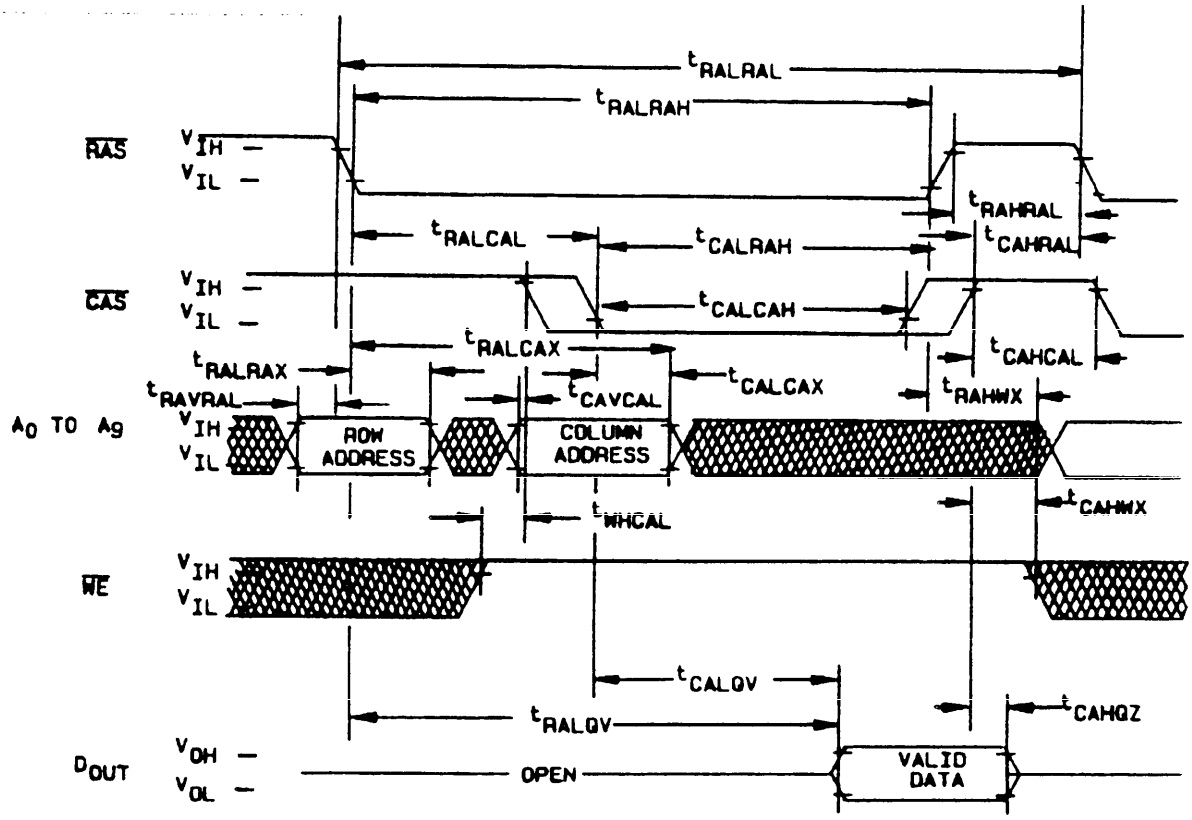


FIGURE 8. Read cycle waveforms.

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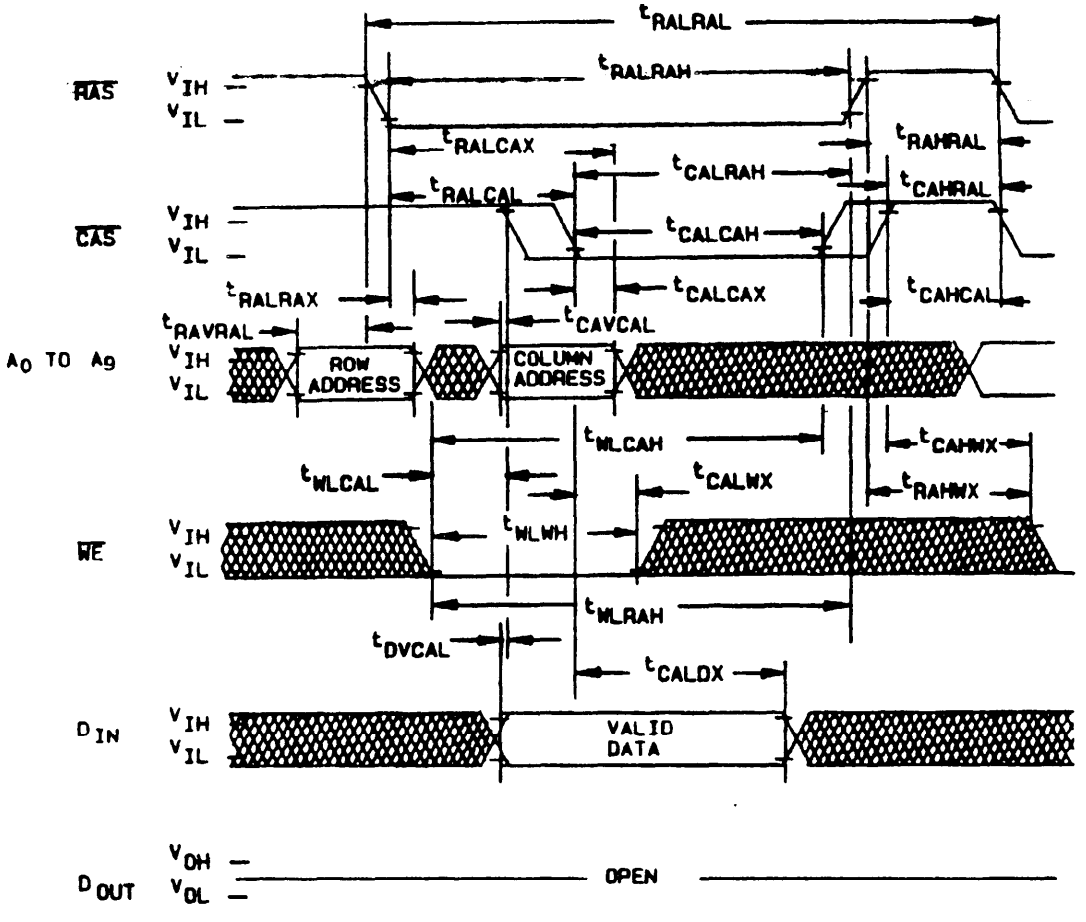


FIGURE 9. Write cycle waveforms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:58Z
Check the source to verify that this is the current version before use.

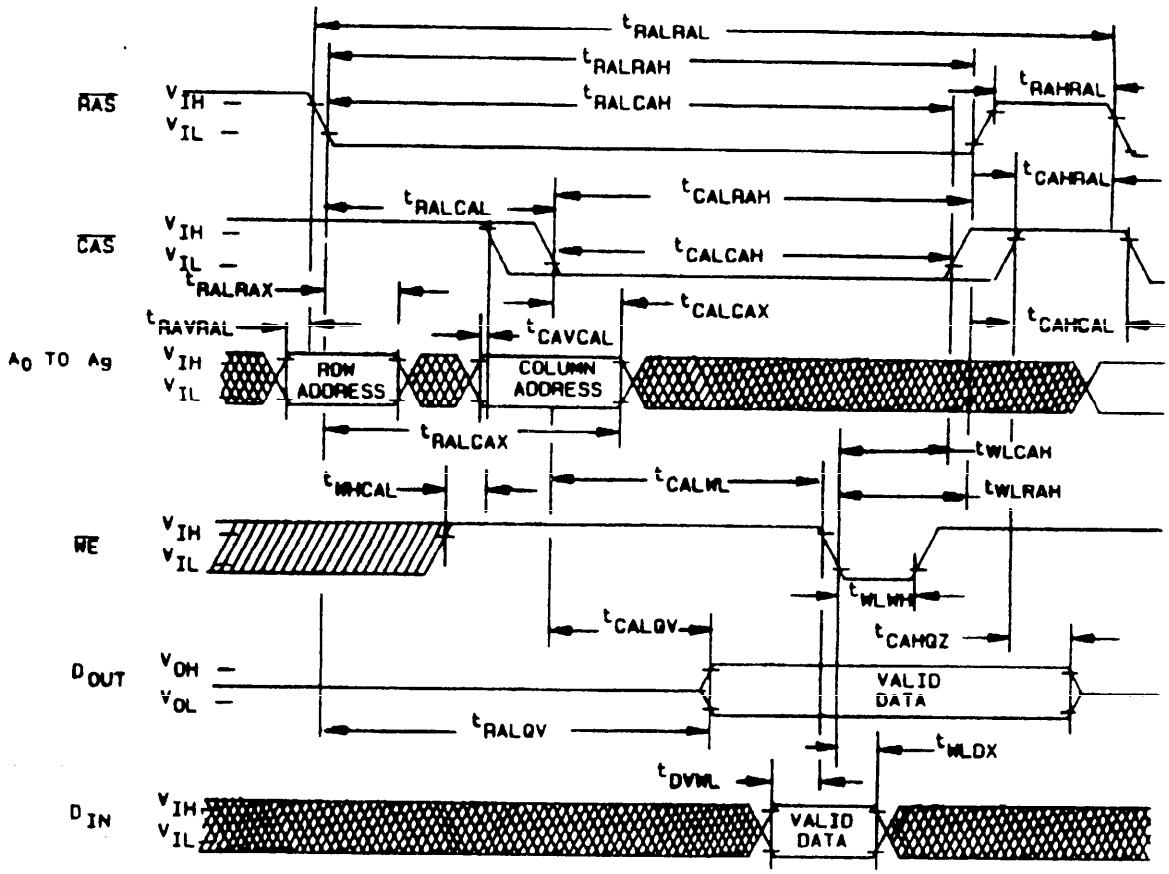
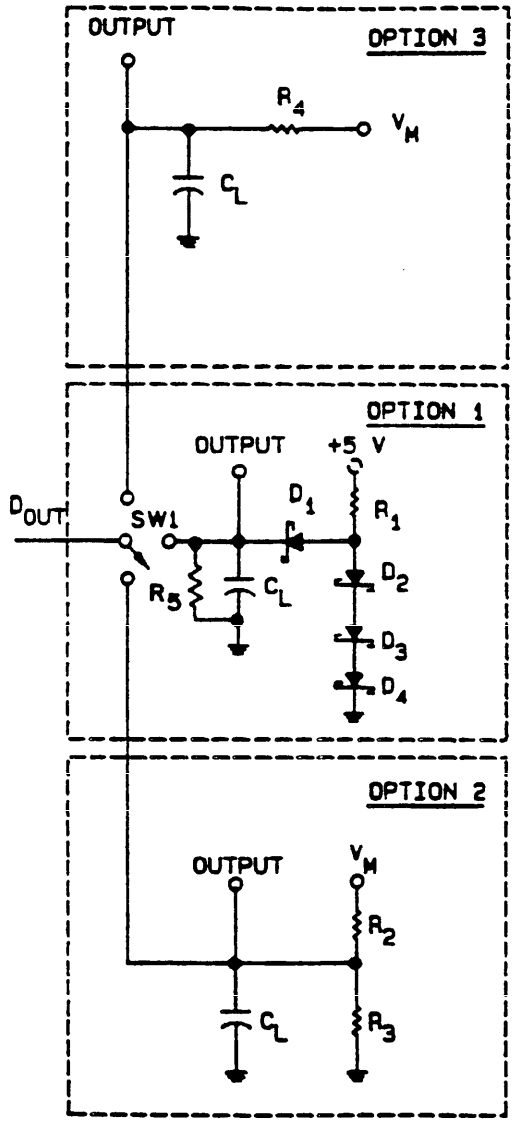


FIGURE 10. Read/write, read/modify/write cycle waveforms.

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NOTES:
1. C_L = 50 pF minimum (includes scope probe, wiring, and stray wiring without package on test fixture).
2. D_1-D_4 are 1N3064 or equivalent.
3. Option 2 conditions are: V_M = 5.0 V, R_2 = 910Ω ±5 percent, R_3 = 320Ω ±5 percent.
4. All resistors are 1/2 watt.
5. Options 1, 2, or 3 load circuit may be used for all ac tests except the high impedance test, where option 2 shall be used.
6. SW1 is a software switch.
7. Option 1 conditions are: R_1 = 1 kΩ ±1 percent, R_5 = 480Ω ±1 percent.
8. Option 3 conditions are: V_M = 1.3 V, R_4 = 220Ω ±5 percent.

FIGURE 11. Switching time test circuit or equivalent.

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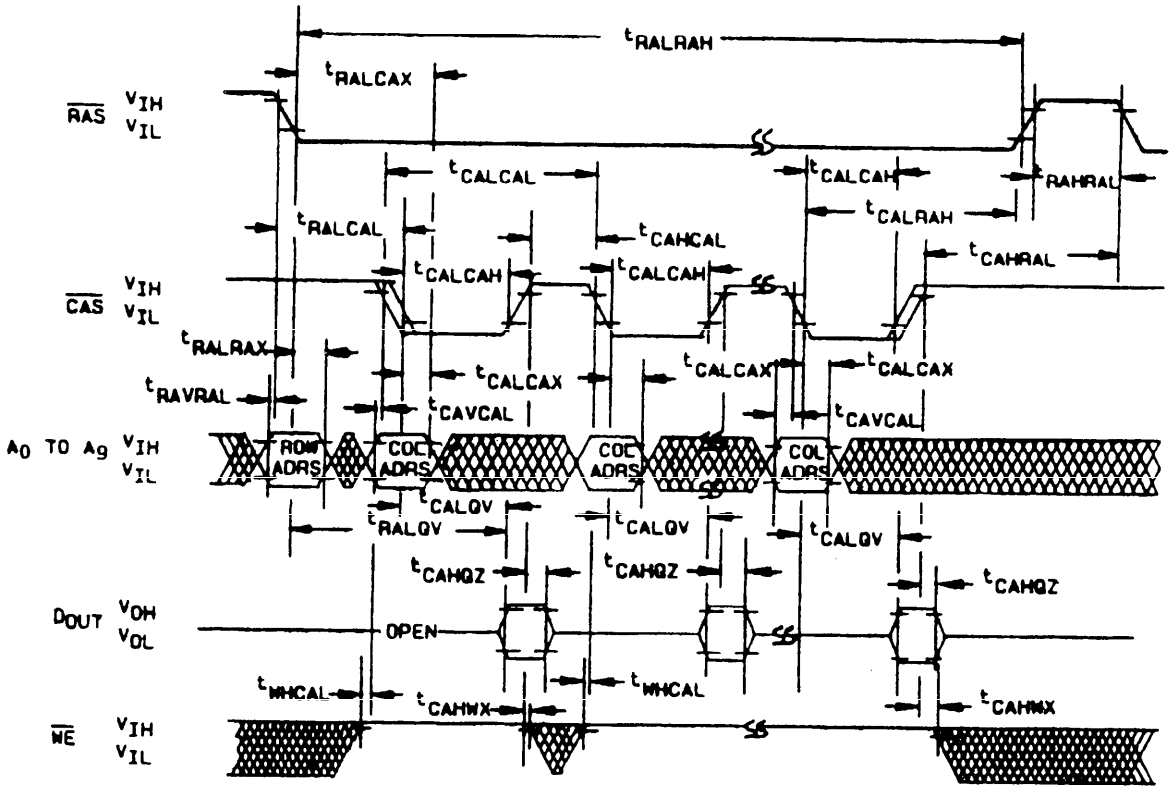


FIGURE 12. Page mode read cycle waveforms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:58Z
Check the source to verify that this is the current version before use.

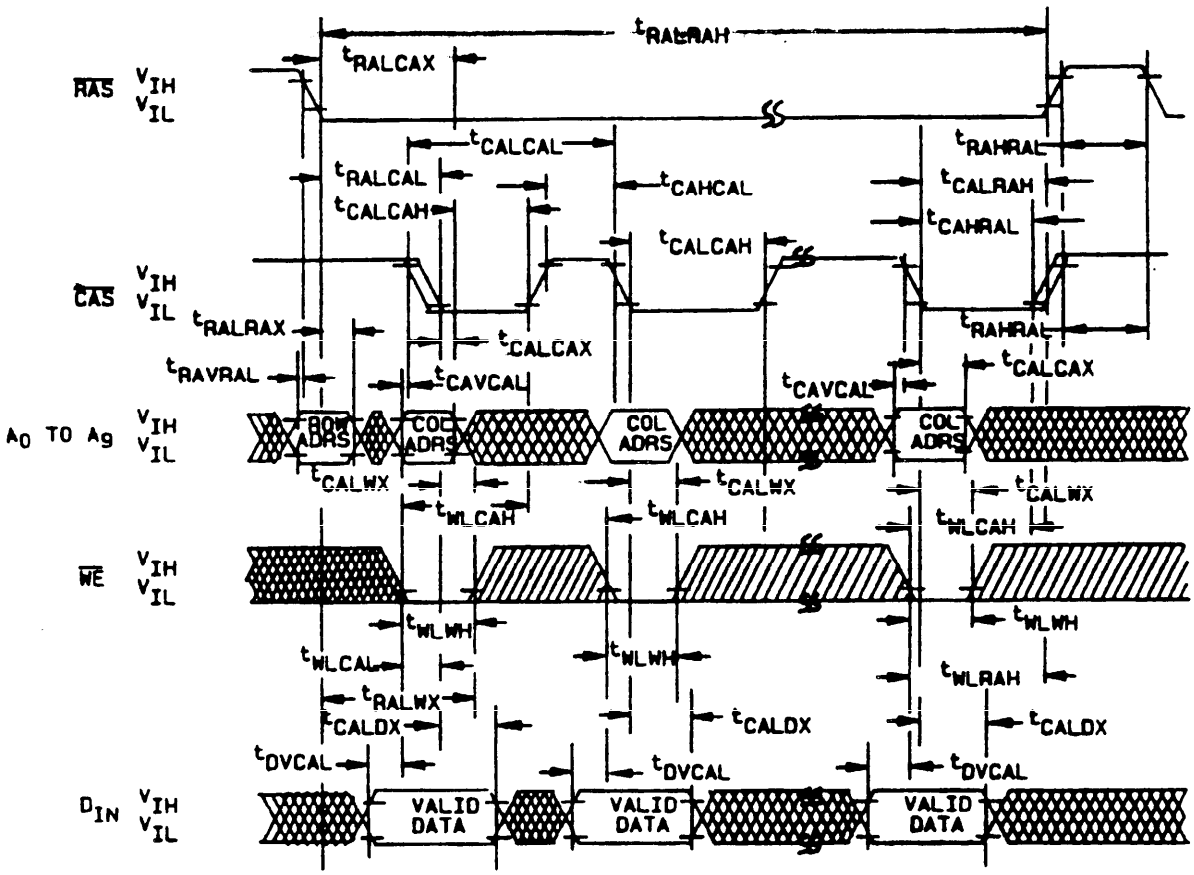


FIGURE 13. Page mode write cycle waveforms.

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TABLE III. Group A inspection for device types 01, 02, 03, and 04.
Terminal conditions: Pins not designated may be high ≥ 2.4 V, or low ≤ 0.8 V, or open.

Subgroup	Symbol	MIL-STD-883 method	Cases Y, Z	1	2	3	9	10	11	12	13	14	15	16	17	18	22	24	25	26	Measured terminal	Test limits		Unit
			Cases X, U	1	2	3	6	7	8	9	10	11	12	13	14	15	16	18	19	20				
			Case V	1	2	3	5	6	7	8	9	10	11	12	13	14	15	16	17	18				
			Test no.	D_{IN}	$\overline{WE}$	$\overline{RAS}$	A_0	A_1	A_2	A_3	V_{CC}	A_4	A_5	A_6	A_7	A_8	A_9	$\overline{CAS}$	D_{OUT}	V_{SS}		Min	Max	
1 T_C = +25°C	I_{CC1}	3005	1	1/	1/	1/	1/	1/	1/	1/	5.5 V	1/	1/	1/	1/	1/	1/	1/	2/	0.0 V	V_{CC}		3/	mA
	I_{CC2}	3005	2	0.0 V	2.4 V	2.4 V	0.0 V	0.0 V	0.0 V	0.0 V	"	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	2.4 V	0.0 V	"	"		"	"
	I_{CC3}	3005	3	4/	4/	4/	4/	4/	4/	4/	"	4/	4/	4/	4/	4/	4/	2.4 V	4/	"	"		"	"
	I_{CC4}	3005	4	5/	5/	5/	5/	5/	5/	5/	"	5/	5/	5/	5/	5/	5/	5/	5/	"	"		"	"
	V_{OH}	3006	5	6/	6/	6/	6/	6/	6/	6/	4.5 V	6/	6/	6/	6/	6/	6/	6/	-5 mA	"	D_{OUT}	2.4		V
	V_{OL}	3006	6	6/	6/	6/	6/	6/	6/	6/	"	6/	6/	6/	6/	6/	6/	6/	4 mA	"	"		0.4	"
	V_{OZ}		7	7/	7/	7/	7/	7/	7/	7/	"	7/	7/	7/	7/	7/	7/	7/	7/	"	"	8/	8/	"
	I_{IH}	3010	8	6.5 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	5.5 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	"	D_{IN}	-10	+10	µA
			9	0.0 V	6.5 V	0.0 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	$\overline{WE}$	"	"	"
			10	"	0.0 V	6.5 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	$\overline{RAS}$	"	"	"
			11	"	"	0.0 V	6.5 V	"	"	"	"	"	"	"	"	"	"	"	"	"	A_0	"	"	"
			12	"	"	"	0.0 V	6.5 V	"	"	"	"	"	"	"	"	"	"	"	"	A_1	"	"	"
			13	"	"	"	"	0.0 V	6.5 V	"	"	"	"	"	"	"	"	"	"	"	A_2	"	"	"
			14	"	"	"	"	"	0.0 V	6.5 V	"	"	"	"	"	"	"	"	"	"	A_3	"	"	"
			15	"	"	"	"	"	"	0.0 V	"	6.5 V	"	"	"	"	"	"	"	"	A_4	"	"	"
			16	"	"	"	"	"	"	"	"	0.0 V	6.5 V	"	"	"	"	"	"	"	A_5	"	"	"
			17	"	"	"	"	"	"	"	"	"	0.0 V	6.5 V	"	"	"	"	"	"	A_6	"	"	"
			18	"	"	"	"	"	"	"	"	"	"	0.0 V	6.5 V	"	"	"	"	"	A_7	"	"	"
			19	"	"	"	"	"	"	"	"	"	"	"	0.0 V	6.5 V	"	"	"	"	A_8	"	"	"
			20	"	"	"	"	"	"	"	"	"	"	"	"	0.0 V	6.5 V	"	"	"	A_9	"	"	"
			21	"	"	"	"	"	"	"	"	"	"	"	"	0.0 V	0.0 V	6.5 V	"	"	$\overline{CAS}$	"	"	"
	I_{IL}	3009	22	"	6.5 V	6.5 V	6.5 V	6.5 V	6.5 V	6.5 V	"	6.5 V	6.5 V	6.5 V	6.5 V	6.5 V	6.5 V	6.5 V	6.5 V	"	D_{IN}	"	"	"
			23	6.5 V	0.0 V	6.5 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	$\overline{WE}$	"	"	"
			24	"	6.5 V	0.0 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	$\overline{RAS}$	"	"	"
			25	"	"	6.5 V	0.0 V	"	"	"	"	"	"	"	"	"	"	"	"	"	A_0	"	"	"
			26	"	"	"	6.5 V	0.0 V	"	"	"	"	"	"	"	"	"	"	"	"	A_1	"	"	"
			27	"	"	"	"	6.5 V	0.0 V	"	"	"	"	"	"	"	"	"	"	"	A_2	"	"	"
			28	"	"	"	"	"	6.5 V	0.0 V	"	"	"	"	"	"	"	"	"	"	A_3	"	"	"
			29	"	"	"	"	"	"	6.5 V	"	0.0 V	"	"	"	"	"	"	"	"	A_4	"	"	"
			30	"	"	"	"	"	"	"	"	6.5 V	0.0 V	"	"	"	"	"	"	"	A_5	"	"	"
			31	"	"	"	"	"	"	"	"	"	6.5 V	0.0 V	"	"	"	"	"	"	A_6	"	"	"
			32	"	"	"	"	"	"	"	"	"	"	6.5 V	0.0 V	"	"	"	"	"	A_7	"	"	"
			33	"	"	"	"	"	"	"	"	"	"	"	6.5 V	0.0 V	"	"	"	"	A_8	"	"	"
			34	"	"	"	"	"	"	"	"	"	"	"	"	6.5 V	0.0 V	"	"	"	A_9	"	"	"
			35	"	"	"	"	"	"	"	"	"	"	"	"	6.5 V	6.5 V	0.0 V	"	"	$\overline{CAS}$	"	"	"
	I_{OHZ} 7/	3021	36	0.0 V	2.4 V	2.4 V	0.0 V	0.0 V	0.0 V	0.0 V	"	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	2.4 V	"	"	D_{OUT}	"	"	"
	I_{OLZ} 7/	3020	37	0.8 V	6.5 V	6.5 V	0.8 V	0.8 V	0.8 V	0.8 V	"	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	6.5 V	0.0 V	"	D_{OUT}	"	"	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C																							
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C																							

See footnotes at end of table III.

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MIL-M-38510/49A

TABLE III. Group A inspection for device types 01, 02, 03, and 04 - Continued.
Terminal conditions: Pins not designated may be high ≥ 2.4 V, or low ≤ 0.8 V, or open.

Subgroup	Symbol	MIL-STD-883 method	Cases Y, Z	1	2	3	9	10	11	12	13	14	15	16	17	18	22	24	25	26	Measured terminal	Test limits		Unit
			Cases X, U	1	2	3	6	7	8	9	10	11	12	13	14	15	16	18	19	20				
			Case V	1	2	3	5	6	7	8	9	10	11	12	13	14	15	16	17	18				
			Test no.	D_{IN}	$\overline{WE}$	$\overline{RAS}$	A_0	A_1	A_2	A_3	V_{CC}	A_4	A_5	A_6	A_7	A_8	A_9	$\overline{CAS}$	D_{OUT}	V_{SS}		Min	Max	
4	C_{IN1} C_{IN2} C_{OUT}	3012	38 39 40	Open 9/ Open	Open 9/ Open	Open 9/ Open	9/ Open Open	9/ Open Open	9/ Open Open	9/ Open Open	5.0 V " "	9/ Open Open	9/ Open Open	9/ Open Open	9/ Open Open	9/ Open Open	9/ Open Open	Open 9/ Open	Open Open 9/	0.0 V " "	A_0-A_9 Clocks D_{OUT}		7 10 8	pF " "
7 T_C = +25°C	I_{CC2L1} I_{CC1L2} I_{CC2L3} Func-tional		41 42 43 44	10/ 11/ 12/	10/ 11/ 12/	10/ 11/ 12/	10/ 11/ 12/	10/ 11/ 12/	10/ 11/ 12/	10/ 11/ 12/	5.5 V " "	10/ 11/ 12/	10/ 11/ 12/	10/ 11/ 12/	10/ 11/ 12/	10/ 11/ 12/	10/ 11/ 12/	10/ 11/ 12/	10/ 11/ 12/	" " "	V_{CC} " "	3/ " "	3/ " "	mA " "
8	Same functional tests (truth table verification) as subgroup 7 (excluding tests 41, 42, and 43, except repeat at T_C = +125°C and T_C = -55°C.																							
9 T_C = +25°C	Table I ac param-eters		45	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	4.5 V	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	0.0 V	D_{OUT}	3/	3/	ns
			46	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	5.5 V	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	3/ 13/ 14/ 15/ 16/ 17/	"	"	"	"	"
			47	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	4.5 V	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	"	"	"	"	"
			48	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	5.5 V	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	3/ 14/ 15/ 16/ 17/	"	"	"	"	"
			49	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	4.5 V	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	"	"	"	"	"
			50	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	5.5 V	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	3/ 14/ 16/ 17/	"	"	"	"	"
			51	3/ 14/ 17/	3/ 14/ 17/	3/ 14/ 17/	3/ 14/ 17/	3/ 14/ 17/	3/ 14/ 17/	3/ 14/ 17/	4.5 V	3/ 14/ 17/	3/ 14/ 17/	3/ 14/ 17/	3/ 14/ 17/	3/ 14/ 17/	3/ 14/ 17/	3/ 14/ 17/	3/ 14/ 17/	"	"	"	"	"
			52	3/ 14/ 17/	3/ 14/ 17/	3/ 14/ 17/	3/ 14/ 17/	3/ 14/ 17/	3/ 14/ 17/	3/ 14/ 17/	5.5 V	3/ 14/ 17/	3/ 14/ 17/	3/ 14/ 17/	3/ 14/ 17/	3/ 14/ 17/	3/ 14/ 17/	3/ 14/ 17/	3/ 14/ 17/	"	"	"	"	"
10	Same tests, terminal conditions, and limits as subgroup 9, except T_C = +125°C																							
11	Same tests, terminal conditions, and limits as subgroup 9, except T_C = -55°C																							

1/ Timing edges are set to minimum limits in accordance with table I for t_{RALRAL}, t_{RALRAH}, t_{CALCAH}, t_{WLCAL}.
2/ Output load I_{OH} or I_{OL} of table I and 50 pF minimum to V_{SS} or equivalent (see figure 11).
3/ Multiple timings sets shall be used so as to test all minimum and maximum timing parameters specified in table I. These shall be performed using the appropriate pattern as described in 3.5.1.
4/ Timing edges are set to minimum limits in accordance with table I for t_{RALRAL}, t_{RALRAH}.
5/ Timing edges are set to minimum limits in accordance with table I for t_{CALCAL}, t_{CAHCAL}.
6/ These tests are listed but not tested. The data output conditions and test limits shall be set and used for conditions for go/no-go testing.
7/ Multiple toggles between 2.4 V and 0.8 V are applied to $\overline{RAS}$ and $\overline{CAS}$; after toggling $\overline{RAS}$ and $\overline{CAS}$ are 2.4 V.
8/ Measure output voltage to be > V_{OL} +0.5 V for a "0", and < V_{OH} -0.5 V for a "1" after t_{CAHQX} maximum delay.
9/ Capacitance measured with Boonton 72BD or equivalent with V_{bias} = 2.4 V applied to each input and output individually with remaining input and output pins open.
10/ Static supply current latchup test. All input and output pins shall withstand -1.0 V and V_{CC} + 1 V for 10 ms without exceeding specified I_{CC} limit of table I (see 4.4.1d).
11/ Dynamic supply current latchup test. All inputs swing between -1.5 V and 7.0 V at 500 ns cycle time without exceeding I_{CC} limit of table I (see 4.4.1d). Data output open.
12/ Static supply current latchup test. All inputs and data output equal V_{CC} +1 V; V_{CC} ramps from 0 V to 5 V; ramp rate less than 5 ms/V without exceeding I_{CC} limit of table I (see 4.4.1d).
13/ Input levels are 2.4 V and 0.8 V. Output levels are 2.4 V and 0.4 V.
14/ Input and output levels from footnotes 13/, 15/, 16/, and 17/ are tested using only the marching columns algorithm J. All other algorithms are tested with the levels in footnote 13/ only.
15/ Input levels are 2.4 V and -1.0 V. Output levels are 2.4 V and 0.4 V.
16/ Input levels are 6.5 V and -1.0 V. Output levels are 2.4 V and 0.4 V.
17/ Input levels are 6.5 V and 0.8 V. Output levels are 2.4 V and 0.4 V.

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4.3.2 Electrostatic discharge sensitivity qualification inspection. Electrostatic discharge sensitivity (ESDS) testing shall be performed in accordance with MIL-STD-883, method 3015. The option to categorize devices as ESD sensitive without performing the test is not allowed. Only those device types that pass ESDS testing at 1,000 volts or greater shall be considered as conforming to the requirements of this specification. ESDS testing shall be measured only for initial qualification and after process or design changes which may affect ESDS classification.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical tests shall be as specified in table II herein.

b. Subgroups 5 and 6 for class S and subgroups 5 and 6 for class B of table I of method 5005 of MIL-STD-883 shall be omitted.

c. Subgroup 4 shall be measured only for initial qualification and after process or design changes which may affect capacitance. Sample size is fifteen devices with no failures, and all input and output terminals tested. Perform C_{IN1}, C_{IN2}, and C_{OUT} parameter measurements to table I limits.

d. I_{CC2L1}, I_{CC1L2}, and I_{CC2L3} of subgroup 7 shall be measured only for initial qualification and after process or design changes which may affect these three parameters. Latchup test shall be considered destructive. Test all applicable pins on 5 devices with no failures. Equivalent latch up test procedures shall be defined during initial qualification and submitted to the qualifying activity for approval.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Class S electrical test requirements shall be as specified in table II herein. Delta limits shall apply only to subgroup 5 of group B inspection and shall consist of tests specified in table IV herein.

b. Steady-state life test for class S devices shall be in accordance with table IIa (subgroup 5) of method 5005 of MIL-STD-883, using a circuit submitted to the qualifying activity for approval. If the alternate burn-in conditions are used, the circuit on figure 5 or equivalent shall be used.

TABLE IV. Delta limits at +25°C.

Parameter 1/	Limits 01, 02, 03, 04
I_{CC2}	±0.1 mA
I_{OZ}	±50 nA
I_{IL}, I_{IH}	±50 nA

1/ The above parameters shall be recorded before and after the required burn-in and life tests to determine deltas (Δ).

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4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical tests shall be as specified in table II herein. Delta limits shall apply only to subgroup I of group C inspection and shall consist of tests specified in table IV herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 5, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

c. Additional group C subgroup (C-3). Alpha particle soft error rate testing shall be performed in accordance with method 1032 of MIL-STD-883. Testing shall be performed only for initial qualification and after process, design or materials that may affect soft error susceptibility. The failure rate shall be as specified in 1.4. System testing to be performed using conditions stated in 1.4. Accumulate 10^6 device hours minimum for the specified generic device type. As an alternate, a manufacturer may submit a plan that contains an accelerated technique that can be demonstrated to show that the soft error rate requirement will be met.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883 and as follows. End-point electrical parameters shall be as specified in table II herein. For those designs using a die coat, a measurement shall be made of the thickness of the die coat. The thickness shall be greater than or equal to that measured on the initial qualification devices or the devices used for requalification after a change (see 4.4.1c). If a die overcoat is applied to the wafer fabrication then the thickness is controlled by the applicable wafer fabrication baseline and no measurement is required at assembly.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit V_{SS} terminal. Currents given are conventional and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cool down procedures. When the burn-in and life tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to within 10°C of their power stable condition at room temperature; then, electrical parameter end-point measurements shall be performed.

4.5.3 Delta measurements. Delta measurements, as specified in table II, shall be made before and after the required burn-in screens and steady-state life tests to determine delta compliance. The electrical parameters to be measured with associated delta limits are listed in table IV.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

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6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for use for Government microcircuit applications (original equipment) and logistic purposes.

6.2 Ordering data. Acquisition documents should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

V_{CC}	Supply voltage
V_{SS}	Common or reference voltage
$\overline{RAS}$	Row enable input
$\overline{CAS}$	Column enable input
$\overline{WE}$	Write/read input
D_{IN}	Data input
D_{OUT}	Data output
A_0-A_9, ADDR, ADRS	Address input
LSB	Least significant address bit
MSB	Most significant address bit
NC	No connect
TS no.	Timing set number

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6.4 Application and operation guidelines.

6.4.1 Initialization. The device requires a 100 µs pause and eight pump cycles after power up before it will operate properly. Any cycle that performs refresh may be used for a pump cycle.

6.4.2 Addressing. Addressing 1 of 1,048,576 (2^{20}) cells requires handling a 20-bit address word. This is accomplished with minimum pin count by multiplexing two 10-bit address fields onto the ten address inputs (A_0-A_9). The two fields are brought on chip in succession by the high to low transition of row enable and column enable ($\overline{RAS}$, $\overline{CAS}$) which activate internal timing generators and latch the address information on chip for decoding.

6.4.3 Read operation. When a valid row address is presented to the address inputs, the row enable ($\overline{RAS}$) can fall and thereby clock the row address into the address latches. After a row address hold time (t_{RALAX}), the column address can be presented to the device and the column enable ($\overline{CAS}$) can fall to latch the column address onto the chip. The $\overline{WE}$ input is held at a logic "1" (high) level. The output is in the high impedance state and will remain in that state until access time. At access time it will turn on and assume the appropriate level ("1" or "0"). A feature referred to as "gated $\overline{CAS}$" is simply a delayed internal signal that is gated with the external $\overline{CAS}$ signal which causes the on chip $\overline{CAS}$ clock to occur at a fixed interval after $\overline{RAS}$. This allows $\overline{CAS}$ to become active (low) any time after row address hold time (t_{RALRAX}) has been satisfied but before the maximum $\overline{RAS}$ to $\overline{CAS}$ delay time (t_{RALCAL}) without affecting access time referenced to $\overline{RAS}$ (t_{RALQV}). $\overline{CAS}$ can occur later than t_{RALCAL} maximum without affecting device operation but access time will be controlled exclusively by $\overline{CAS}$ which is access time referenced to $\overline{CAS}$ (t_{CALQV}).

6.4.4 Write operation. The same procedure applies for latching the address information on chip that was explained for the read operation. Data can be written into the device in several ways. In all cases, however, writing is accomplished by the falling edge of $\overline{CAS}$ or $\overline{WE}$ (while $\overline{RAS}$ is low) whichever occurs latest. Data is latched onto the chip.

a. Early write: D_{IN} and $\overline{WE}$ precede the fall of $\overline{CAS}$ by a setup time (t_{DVCAL} and t_{WLCAL}), respectively) and remain valid for a hold time (t_{CALDX} and t_{CALWX}). The setup and hold times are with respect to the falling edge of $\overline{CAS}$. Data out remains in the high impedance state for the entire cycle.

b. Late write: The fall of $\overline{WE}$ occurs after D_{IN} and $\overline{CAS}$ by a specified time ($\overline{CAS}$ to $\overline{WE}$ delay t_{CALWL}). D_{IN} setup and hold times are referenced to the fall of $\overline{WE}$. D_{OUT} will contain data from the selected cell at access time. If the fall of $\overline{WE}$ occurs prior to the $\overline{CAS}$ to $\overline{WE}$ delay limit, the state of D_{OUT} pin is indeterminate.

6.4.5 Refresh operations. The basic device constraint is that no cell shall be refreshed at intervals exceeding 8.0 ms. There are 1024 cells along each row in the storage matrix and there are 1024 such rows. A refresh cycle refreshes 1024 cells in each of two rows simultaneously. A row address 512 cycle "burst" every 8.0 ms or a single "distributed" cycle every 15.625 µs at each of the first 512 row addresses (i.e., A_0-A_8; A_9 is a don't care) will accomplish complete memory refreshing. A refresh cycle can be accomplished in three ways. Any normal read, write, or read-modify-write cycle, or a $\overline{RAS}$-only refresh cycle, or a $\overline{CAS}$ before $\overline{RAS}$ will accomplish refresh. The $\overline{RAS}$-only and $\overline{CAS}$ before $\overline{RAS}$ refresh mode requires less power than a standard (PAP) $\overline{RAS}/\overline{CAS}$ cycle. In addition, the $\overline{CAS}$ before $\overline{RAS}$ mode further simplifies refresh by eliminating the need for external applied address.

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6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01, 02, 03	411024-80, 100, 120
01, 02, 03, 04	4C1024-80, 100, 120, 150

6.7 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. Input protection circuitry has been designed into the device to minimize the effect of this static buildup. However, the following practices are recommended.

a. Devices should be handled on benches with conductive and grounded surfaces.

b. Ground test equipment, tools, and operator.

c. Handling of devices by the leads should be avoided.

d. Devices should be stored in conductive foam or carriers.

e. The use of plastic, rubber, or silk in MOS areas should be avoided.

f. Relative humidity should be maintained above 50 percent, if practical.

6.8 Key word listing.

a. Algorithm, functional.

b. Die overcoat.

c. Latchup, supply current.

d. Refresh.

e. Soft error rate, alpha particle.

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APPENDIX

Functional algorithms and timing sets

10. SCOPE

10.1 Scope. Functional algorithms are test patterns which define the sequence of events used to verify address and data pattern sensitivities of a memory. Since devices from different manufacturers do not necessarily exhibit the same sensitivities, the manufacturer may develop alternate and additional algorithms to test for the sensitivities particular to that device. Algorithms shall be applied to the device in a topologically pure fashion. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance. If the manufacturer has built-in parallel test modes designed to speed the functional test time required for the 1-megabit DRAM, these modes may be used for all patterns described in this document, with the exception of those patterns intended to test for address related failure modes (patterns 3 and 4). Provided the following conditions are met by the design of the test mode logic:

Condition 1: The bits making up the parallel data word must be organized such that each bit comes from a completely separate internal array. This condition allows complete pattern sensitivity testing of each individual array in parallel.

Condition 2: The topological address scrambling of each internal array must be such that true surround by complement data patterns can be written to each array simultaneously. This condition, in conjunction with condition 1, guarantees that all fault conditions (except address uniqueness faults) can be tested as thoroughly in the test modes as in normal access mode. Failure to meet conditions 1 and 2 will result in some internal arrays containing the correct data patterns and some internal arrays containing incorrect data patterns while in the test modes.

Condition 3: An error shall be detected by an open output during the read cycle. That is, if all bits are written to a "1" then a "1" shall appear on the data-out pin if all bits read are a "1". Likewise for a "0", if all bits are written to a "0", then a "0" shall appear on the data-out pin if all bits read are a "0". In either case, if one or more of the bits in the test mode word are in the wrong state, the data-out pin shall remain in a high impedance state throughout the read cycle.

10.1.1 Algorithms, step 1. Step 1 of all algorithms may be performed initially and if testing is continuous (no dead time exceeding 8 ms between tests), step 1 does not have to be repeated for each algorithm.

20. REFERENCED DOCUMENTS. This section is not applicable to this appendix.

30. ALGORITHMS

30.1 Algorithm A (pattern 1).

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30.1.1 Output high impedance (t_{CAHQZ}).

This pattern verifies the output buffer switches to high impedance (three-state) within the specified 35 ns after the rise of CAS. It is performed in the following manner:

Step 1	Perform 8 pump cycles.
Step 2	Load minimum address location with "0".
Step 3	Read minimum address location and measure V_{OL}.
Step 4	Raise $\overline{CAS}$ and measure $V_{OUT} \geq V_{OL}$ +0.5 V after 35 ns delay.
Step 5	Load minimum address location with "1".
Step 6	Read minimum address location and measure V_{OH}.
Step 7	Raise $\overline{CAS}$ and measure $V_{OUT} \leq V_{OH}$ -0.5 V after 35 ns delay.

30.2 Algorithm B (pattern 2).

30.2.1 V_{bump}/V_{bobble} data background = all "0" (discharged state).

This pattern indicates sense amplifier margin by slewing the supply voltage between memory writing and reading. It is performed in the following manner:

Step 1	Perform 8 pump cycles.
Step 2	Pause 40 µs ramping V_{CC} to 4.5 V inhibiting all clocks.
Step 3	Load memory with background data.
Step 4	Pause 40 µs ramping V_{CC} to 5.5 V inhibiting all clocks.
Step 5	Read memory with background data.
Step 6	Repeat steps 2 through 5 for background data complement.
Step 7	Load memory with background data.
Step 8	Pause 40 µs ramping V_{CC} to 5.5 V inhibiting all clocks.
Step 9	Read memory with background data.
Step 10	Pause 40 µs ramping V_{CC} to 4.5 V inhibiting all clocks.
Step 11	Repeat steps 7 through 9 for background data complement.

30.3 Algorithm C (pattern 3).

30.3.1 Address complement, data background = all "0".

This pattern produces maximum address line noise and checks the decoder dynamic response times. It is performed in the following manner:

Step 1	Perform 8 pump cycles.
Step 2	Load memory with background data.
Step 3	Read minimum address location.
Step 4	Load minimum address location with "1".
Step 5	Read maximum address location.
Step 6	Load maximum address location with "1".
Step 7	Read minimum address location +1.
Step 8	Load minimum address location +1 with "1".
Step 9	Read maximum address location -1.
Step 10	Load maximum address location -1 with "1".
Step 11	Repeat steps 3 through 10 until all address locations have been read and loaded with "1"s.
Step 12	Repeat steps 3 through 11 reading "1"s and loading "0"s.
Step 13	Read memory, all "0"s.

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30.4 Algorithm D (pattern 4).

30.4.1 March data, data background = all "0".

This pattern tests for address uniqueness and multiple selection. It is performed in the following manner:

Step 1	Perform 8 pump cycles.
Step 2	Load memory with background data.
Step 3	Read location 0.
Step 4	Write data complement in location 0.
Step 5	Read data complement in location 0.
Step 6	Repeat steps 3 through 5 for all other locations in memory (sequentially).
Step 7	Read data complement at minimum location.
Step 8	Write data at minimum location.
Step 9	Read data at minimum location.
Step 10	Repeat steps 7 through 9 for all locations in the memory (sequentially).
Step 11	Repeat steps 2 through 10 with background data of all "1".

30.5 Algorithm E (pattern 5).

30.5.1 Static refresh (periphery retention).

This pattern tests for periphery retention time by attempting to write after a lengthy pause. This test is performed at +125°C (case) only, and is not used to measure the retention time of the periphery circuits, but to insure that they will hold for at least 8 ms. It is performed in the following manner:

Step 1	Perform 8 pump cycles.
Step 2	Load memory with all "0"s.
Step 3	Read memory, all "0"s.
Step 4	Pause (stop all clocks) 8 ms.
Step 5	Load memory with all "1"s.
Step 6	Read memory, all "1"s.
Step 7	Pause (stop all clocks) 8 ms.
Step 8	Load memory with all "0"s.
Step 9	Read memory, all "0"s.

30.6 Algorithm F (pattern 6).

30.6.1 Refresh test (cell retention) +125°C only.

This test is used to check the retention time of memory cells under static and dynamic conditions. It is performed in the following manner:

Step 1	Perform 8 pump cycles.
Step 2	Load memory with all "0"s.
Step 3	Pause 8 ms (stop all clocks).
Step 4	Read memory, all "0"s.
Step 5	Repeat steps 2 through 4 with all "1"s.
Step 6	Read minimum address location.
Step 7	Read maximum address location.
Step 8	Repeat steps 6 and 7 for 8 ms.
Step 9	Read memory, all "1"s.
Step 10	Load memory, all "0"s.
Step 11	Repeat steps 6 through 9 with all "0"s.
Step 12	Read address location 524287.

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Step 13 Read address location 524288.
Step 14 Repeat steps 12 and 13 for 8 ms.
Step 15 Read memory, all "0"s.
Step 16 Load memory, all "1"s.
Step 17 Repeat steps 12 through 15 with all "1"s.

30.7 Algorithm G (pattern 7).

30.7.1 Extended cycle test (10 µs), data background = X-bar.

This test is used to verify the 10 µs maximum limit on $\overline{RAS}$ or $\overline{CAS}$ pulse widths. Front and back edge timing is held to normal cycle timing while the cycle is increased to allow 10 µs of $\overline{RAS}$ and $\overline{CAS}$ active time (low level). It is performed in the following manner:

Step 1 Perform 8 pump cycles.
Step 2 Write data in location 0.
Step 3 Read data in location 0.
Step 4 Repeat steps 2 and 3 for all other locations in the memory (sequentially).
Step 5 Repeat steps 2 through 4 with complement data.

30.8 Algorithm H (pattern 8).

30.8.1 $\overline{RAS}$-only refresh test.

This test is used to verify the functionality of the $\overline{RAS}$-only mode of cell refreshing. It is done at +125°C only and is performed in the following manner:

Step 1 Perform 8 pump cycles.
Step 2 Load memory with all "0"s.
Step 3 Pause 8 ms ($\overline{CAS}$ = $\overline{RAS}$ = $\overline{WE}$ = V_{IH}).
Step 4 Perform 512 $\overline{RAS}$-only cycles ($\overline{CAS}$ = $\overline{WE}$ = V_{IH}).
Step 5 Repeat steps 3 and 4 for 250 ms.
Step 6 Read memory, all "0"s.
Step 7 Repeat steps 2 through 6 with all "1"s.

30.9 Algorithm I (pattern 9).

30.9.1 Read-modify-write (RMW), data background = all "0".

This pattern verifies the read-modify-write mode for the memory. It is performed in the following manner:

Step 1 Perform 8 pump cycles.
Step 2 Load memory with background data.
Step 3 Read minimum address location and load with "1" using RMW cycle.
Step 4 Read maximum address location and load with "1" using RMW cycle.
Step 5 Read minimum address location +1 and load with "1" using RMW cycle.
Step 6 Read maximum address location -1 and load with "1" using RMW cycle.
Step 7 Repeat steps 3 through 6 until all address locations have been read and loaded with "1".
Step 8 Repeat steps 3 through 7 reading "1"s and loading "0"s.
Step 9 Read memory, all "0"s.

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30.10 Algorithm J (pattern 10).

30.10.1 Marching columns, initial data background = all "0".

This pattern is a good test for sense line imbalance and response plus restore noise in addition to multiple selection. It is performed in the following manner:

Step 1	Perform 8 pump cycles.
Step 2	Load memory with data background.
Step 3	Write test column complement.
Step 4	Read test column complement.
Step 5	Write test columns true.
Step 6	Repeat 3 through 6 for all columns sequentially.
Step 7	Write background data to "1".
Step 8	Write test column complement.
Step 9	Read test column complement.
Step 10	Write test columns true.
Step 11	Repeat 8 th;rough 10 for all columns sequentially.

30.11 Algorithm K (pattern 11).

30.11.1 $\overline{\text{CAS}}$-before-$\overline{\text{RAS}}$ refresh test.

This test is used to verify the functionality of the $\overline{\text{CAS}}$ before $\overline{\text{RAS}}$ mode of cell refreshing. It is done at +125°C only and is performed in the following manner:

Step 1	Perform 8 pump cycles.
Step 2	Load memory with all "0"s.
Step 3	Pause 8 ms ($\overline{\text{CAS}}$ = $\overline{\text{RAS}}$ = $\overline{\text{WE}}$ = V_{IH}).
Step 4	Perform 512 $\overline{\text{CAS}}$-before-$\overline{\text{RAS}}$ cycles.
Step 5	Repeat steps 3 and 4 for 250 ms.
Step 6	Read memory, all "0"s.
Step 7	Repeat steps 2 through 6 with all "1"s.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1204)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-38510/249A	1990 February 13

3. DOCUMENT TITLE: MICROCIRCUITS, DIGITAL, CMOS, 1,048, 576/1-BIT DYNAMIC RANDOM ACCESS MEMORY (DRAM), MONOLITHIC SILICON

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED (YYMMDD)

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